Module 3 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 3, Quiz A: Definitions & notation

Reading: Module 3 Read 1
Project: Project 3 · Canvas: Def & Notation quiz (Module 3)

Work these practice items after Read 1 (Sections 1–3: one categorical, one numerical, numerical × categorical). Sections cover: variable types, hand computation (counts, proportions, IQR, five-number summary), plot identification, and graph image items. (R/tidyverse questions live in the Synthesis quiz.)

Graph application, scatter interpretation, and two-way tables → Quiz B after Read 2.

Section A — Variable types and one-variable summaries

A1. Numerical vs categorical

Classify each as numerical or categorical (and binary if categorical):

a) Penguin flipper length (mm)

b) Island: Biscoe / Dream / Torgersen

c) Sex: male/female

d) Clutch completion: yes/no

e) Bill depth (mm)

A2. Best summary for one numerical variable

For one numerical variable, name one graph and two numeric summaries.

A3. Best summary for one categorical variable

For one categorical variable, name one graph and one numeric summary.

A4. Choosing median or mean by purpose

Your dataset has a few very large values. For which kind of question would you report the median, and for which the mean?

Section B — Spread: range, IQR, and standard deviation

B1. Range

Data: $4, 6, 8, 10, 10, 12, 14, 16$. Compute the range.

B2. Quartiles and IQR

Dataset N: $7, 8, 9, 10, 10, 11, 12, 13$. Find Q_1, Q_3, and IQR.

B3. Five-number summary

List the five numbers in the five-number summary in order.

B4. What a boxplot shows

Describe how a boxplot represents the five-number summary.

B5. SD interpretation

In words, what does the sample standard deviation s measure?

B6. IQR vs SD

When would you prefer IQR over SD to describe spread? Why?

Section C — Hand computation: counts and proportions

C1. Coffee order counts

Coffee shop ($n = 12$ orders): Latte 5, Drip 4, Tea 3. How many Latte orders? What is $\hat{p}$ for Latte?

C2. Survey yes/no

Survey ($n = 20$): Yes 13, No 7. Compute $\hat{p}$ for Yes.

C3. Bird feeder species

Feeder ($n = 30$ visits): Robin 12, Sparrow 11, Other 7. What proportion are Robin?

Section D — Compute each statistic by hand (one list: 7, 8, 9, 10, 10, 11, 12, 13)

D1. Mean by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute the mean $\bar{x}$.

D2. Median by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute the median.

D3. Minimum

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. What is the minimum?

D4. Maximum

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. What is the maximum?

D5. First quartile Q_1

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute Q_1.

D6. Third quartile Q_3

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute Q_3.

D7. IQR by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$ (with $Q_1 = 8.5$, $Q_3 = 11.5$). Compute the IQR.

D8. Sample SD by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$ (mean $= 10$). Compute the sample standard deviation s.

Section E — Plot types: bar, pie, histogram, dot plot, boxplot (verbal)

E1. Bar chart purpose

You have species (Adelie / Chinstrap / Gentoo). Which graph shows counts per level?

E2. Pie chart vs bar

When is a pie chart acceptable for one categorical variable?

E3. Pie slice interpretation

A pie chart shows 30% Latte, 25% Drip, 45% Tea ($n = 200$ orders). How many Tea orders?

E4. Histogram vs bar

Histogram bins a numerical variable; bar chart has one bar per category. Which fits flipper length (mm)?

E5. Dot plot use

When is a dot plot especially helpful for one numerical variable?

E6. Boxplot five-number summary

A boxplot shows min=5, $Q_1 = 10$, median=12, $Q_3 = 20$, max=50. Is the distribution skewed? Which way?

E7. Side-by-side boxplots

You compare body mass across three species. Name the graph.

E8. Histogram skew (verbal)

A histogram has a long right tail with most values piled on the left. Describe shape.

E9. Dot plot vs histogram

You have one numerical variable with $n = 15$ observations and want to see every individual value. Dot plot or histogram?

Section F — Statistics: interpret counts, proportions, center, spread (3 each)

F1. Count interpretation

In a frequency table of species, the count $n = 124$ for Gentoo. What does 124 mean?

F2. Total from counts

Counts: A=8, B=12, C=5. What is n?

F3. Proportion from count

$x = 9$ successes out of $n = 36$. State $\hat{p}$.

F4. Mean vs median choice

Income data with a few very high earners. You want to describe a typical household. Report center with mean or median?

F5. SD interpretation 1

$s = 2$ cm vs $s = 15$ cm for the same variable. Which sample is more spread out?

F6. IQR interpretation

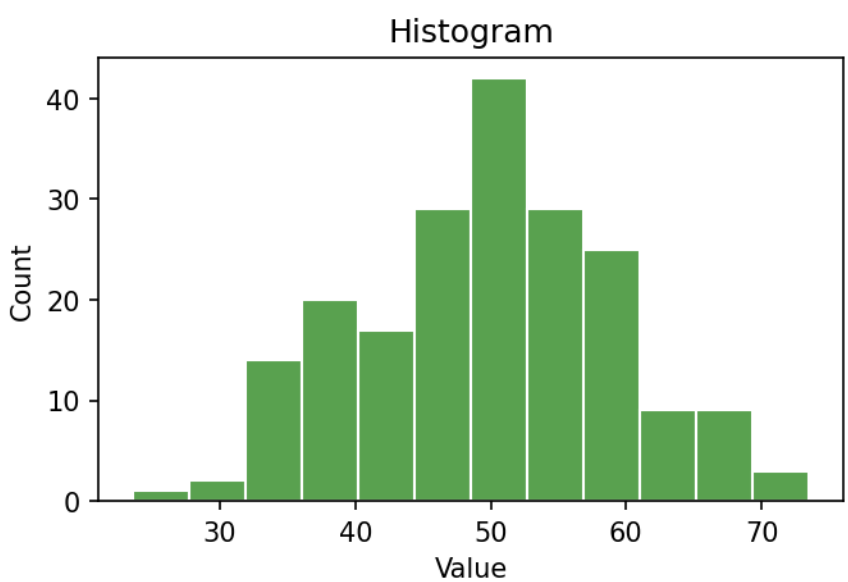
IQR $= 4$ vs IQR $= 20$ for two datasets. Which middle 50% is tighter?

F7. Range sensitivity

Why is range less useful than IQR when outliers exist?

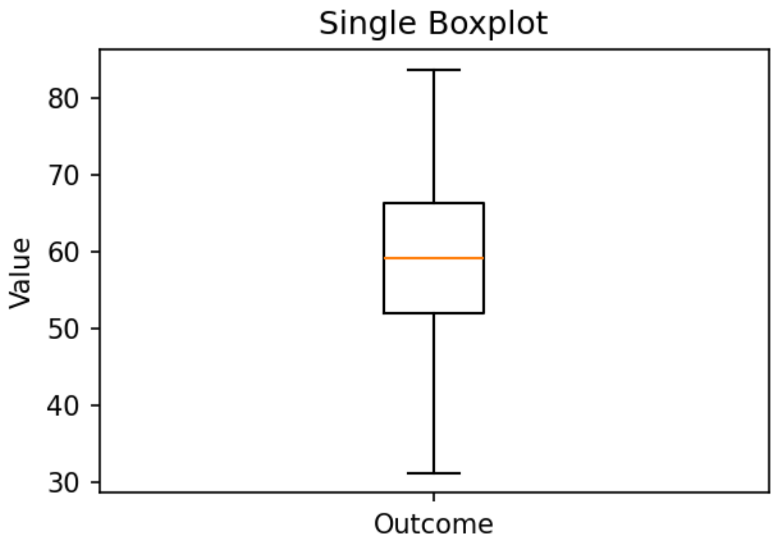
Section G — Graph interpretation (images · Quiz A)

IMG4. Histogram interpretation



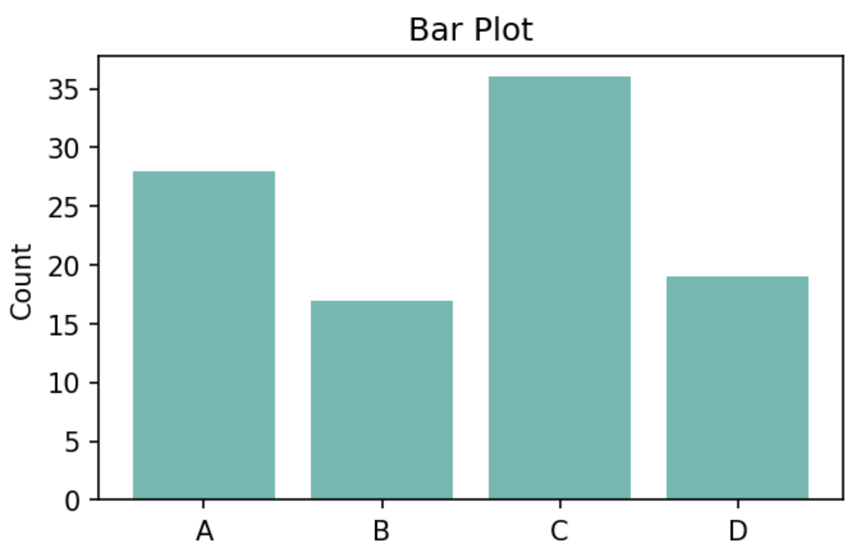
What does this graph primarily display?

IMG6. Single boxplot feature



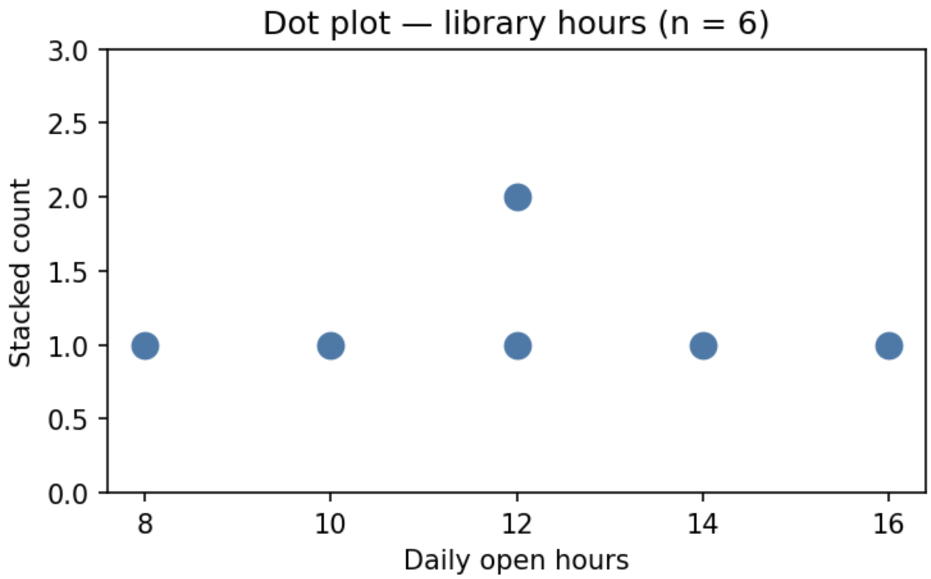
In a boxplot, the horizontal line inside the box marks the:

IMG9. Bar plot variable type



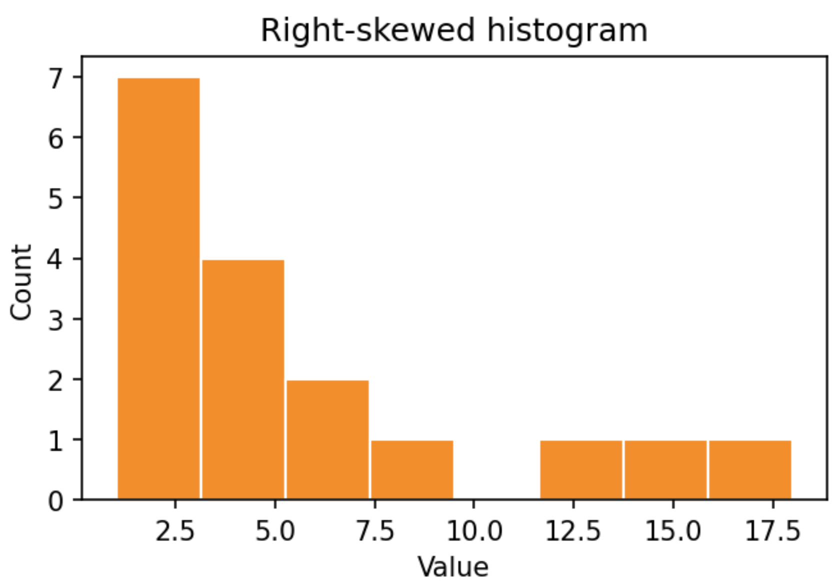
The x-axis categories in a bar plot represent what variable type?

IMG12. Dot plot interpretation



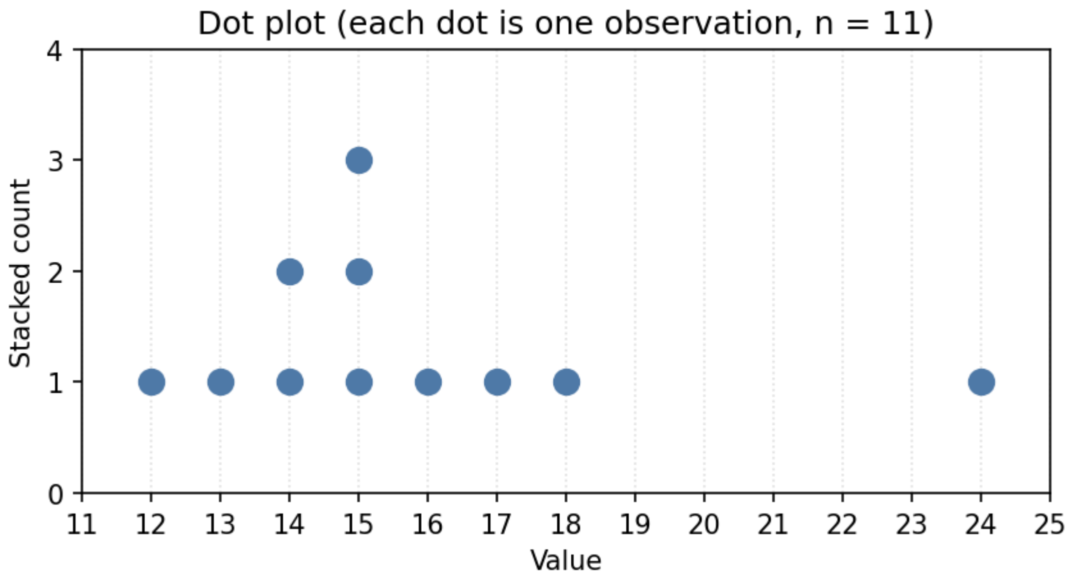
What does this dot plot primarily display?

IMG13. Right-skewed histogram



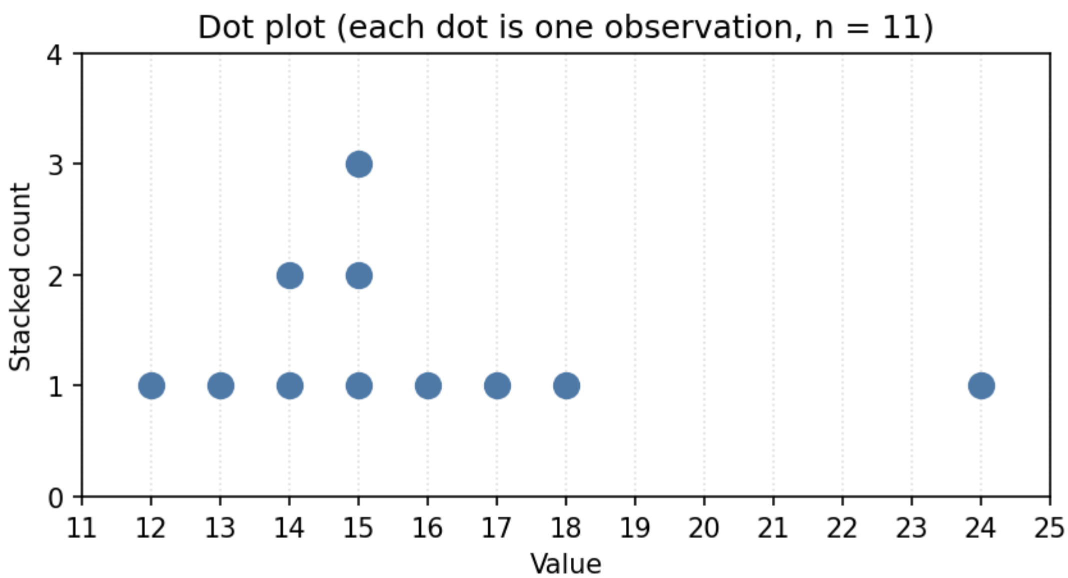
Describe the shape of this histogram.

IMG14. Dot plot — estimate the center



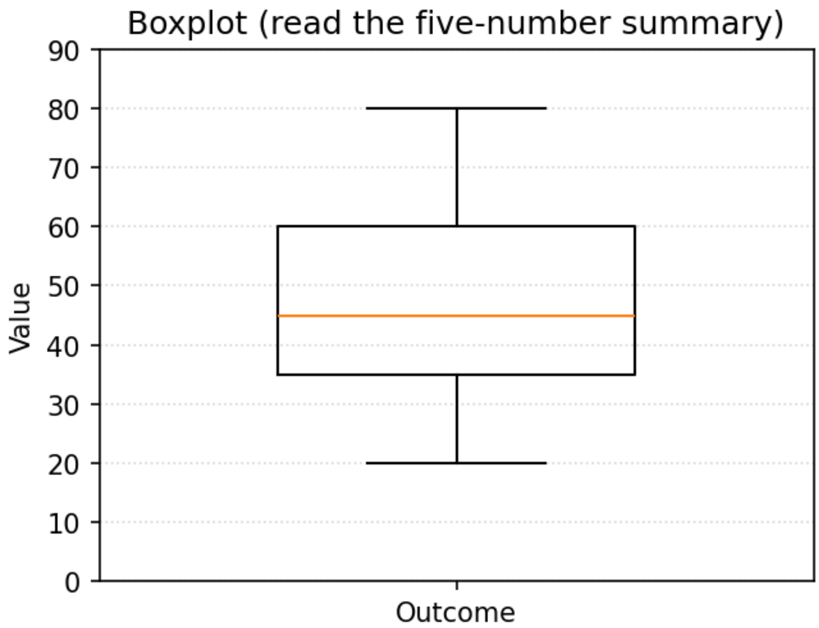
Each dot is one observation. Which value best estimates the center (median) of this dot plot?

IMG15. Dot plot — estimate the range



Estimate the range (largest value minus smallest value) shown in this dot plot.

IMG16. Boxplot — estimate the IQR



Read this boxplot. Estimate the IQR (the height of the box, Q3 − Q1).